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MIL-STD 1568E
11 October 2023
SUPERSEDING
MIL-STD-1568D
31 August 2015

DEPARTMENT OF DEFENSE

DESIGN CRITERIA STANDARD

MATERIALS AND PROCESSES FOR CORROSION PREVENTION AND CONTROL IN AEROSPACE WEAPONS SYSTEMS

AMSC 10432

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocCom.fct@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

3. For guidance on the technical content of this document, contact Commanding Officer, Naval Air Warfare Center, Aircraft Division, Code AB23600, 48066 Shaw Road, 2188 Patuxent River, MD 20670.

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1. SCOPE

1.1 Scope. This standard defines the engineering practices, methods, and procedures for the selection and use of materials and processes required for optimum corrosion prevention and control (CPC) of aerospace weapons systems throughout their life cycle (design, acquisition and sustainment phases).

1.2 Purpose. The purpose of this standard is to provide requirements for implementation of materials, processes and procedures that provide optimum performance, reliability, environmental protection, and life cycle cost for aerospace weapons systems. This standard defines technical requirements to ensure establishment and implementation of a CPC plan and its accompanying finish specification. The CPC plan will set up operating procedures and the finish philosophies used in the systems. The finish specification details the finish and coating systems to be used on the respective aerospace weapons system in accordance with MIL-STD-7179 and the finish philosophies as approved in the corrosion prevention and control plan. Detailed materials and process requirements for aerospace weapons systems are defined in MIL-STD-1587. This standard is derived from experience gained on protection of aerospace weapons systems against corrosion by the military services and industry. Meeting or exceeding the requirements of this standard, as well as applicable program-specific technical documentation, will facilitate an optimum balance between acquisition and sustainment costs for Department of Defense (DoD) aerospace weapons systems, and ultimately enhance system reliability, materials compatibility, supportability, and safety.

1.3 Applicability. This standard is applicable for the DoD procuring activities and their respective contractors involved in the design, procurement and sustainment of aerospace weapons systems, including spares, modifications, upgrades, and major repairs. Applicable systems include fixed wing aircraft, unmanned aerial vehicles, space vehicles, rotorcraft, missile systems, air delivered guided munitions, armament systems and all related ground support equipment. Authority to deviate from the requirements in this standard requires approval from the Cognizant Engineering Authority (CEA).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5002	-	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-PRF-8516	-	Sealing Compound, Synthetic Rubber, Electric Connectors, And Electric Systems, Chemically Cured
MIL-DTL-18264	-	Finishes, Organic, Weapons Systems, Application and Control of
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23586	-	Sealing Compound (With Accelerator), Silicone Rubber, Electrical
MIL-M-24041	-	Molding and Potting Compound, Chemically Cured, Polyurethane
MIL-PRF-27617	-	Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant
MIL-PRF-32014	-	Grease, Aircraft and Instrument
MIL-PRF-32239	-	Coating System, Advanced Performance, for Aerospace Applications
MIL-DTL-32648	-	Zinc Nickel Plating, Low Hydrogen Embrittlement Electro-Deposition
MIL-A-46146	-	Adhesives-Sealants, Silicone, RTV, Noncorrosive (for use with Sensitive Metals and Equipment)
MIL-PRF-46170	-	Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544
MIL-DTL-53072	-	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-PRF-81733	-	Sealing and Coating Compound, Corrosion Inhibitive
MIL-PRF-81322	-	Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code G-395
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-85285	-	Topcoat, Aircraft and Support Equipment
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne
MIL-PRF-87252	-	Coolant Fluid, Hydrolytically Stable, Dielectric
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-186	-	Protective Finishing for Army Missile Weapon Systems Manufacturing Process
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-2073-1	-	Standard Practice for Military Packaging

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MIL-STD-7179	-	Finishes, Coatings, and Sealants, for the Protection of Aerospace Weapons Systems and Support Equipment
MIL-STD-1587	-	Materials and Process Requirements for Aerospace Weapon Systems

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-275	-	Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components
MIL-HDBK-838	-	Lubrication of Military Equipment

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)
TT-P-2760	-	Primer Coating: Polyurethane, Elastomeric, High Solids

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2. Other Government documents, drawings, and publications. Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE DIRECTIVE

DoDD 5000.01	The Defense Acquisition System

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

GUIDEBOOK

Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment

(This tool is available at https://www.dau.edu/tools/t/CPC-Guidebook.)

DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT (DFARS)

Subpart 207.105	-	Acquisition Planning: Contents of Written Acquisition Plans

(Copies of this document are available at https://www.dcaa.mil/dfars.html.)

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JOINT SERVICE SPECIFICATION GUIDE

JSSG-2007	-	Engines, Aircraft, Turbine

(Copies of this document are available at https://www.quicksearch.dla.mil.)

MARSHALL SPACE FLIGHT CENTER (NASA)

MSFC-STD-3029	-	Guidelines for the Selection of Metallic Materials for Stress Corrosion Cracking Resistance in Sodium Chloride Environments

(Copies of this document are available at https://www.nasa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments
ASTM F945	-	Standard Test Method for Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials
ASTM G39	-	Standard Practice for Preparation and Use of Bent-Beam Stress-Corrosion Test Specimens
ASTM G44	-	Standard Practice for Exposure of Metals and Alloys by Alternate Immersion in Neutral 3.5% Sodium Chloride Solution
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products
ASTM G49	-	Standard Practice for Preparation and Use of Direct Tension Stress-Corrosion Test Specimens
ASTM G58	-	Standard Practice for Preparation of Stress-Corrosion Test Specimens for Weldments
ASTM G64	-	Standard Classification of Resistance to Stress-Corrosion Cracking of Heat-Treatable Aluminum Alloys
ASTM G85	-	Standard Practice for Modified Salt Spray (Fog) Testing

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement

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SAE AMS1389	-	Sealing Compound, Polythioether Rubber, Light-Cured, for Integral Fuel Tanks and General Purpose, Intermittent use to 360 ºF (182 ºC)
SAE AMS2430	-	Shot Peening
SAE AMS2432	-	Shot Peening, Computer Monitored
SAE AMS2488	-	Anodic Treatment of Titanium and Titanium Alloys Solution pH 13 or Higher
SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AMS2759/9	-	Hydrogen Embrittlement Relief (Baking) of Steel Parts
SAE AMS3265	-	Sealing Compound, Polysulfide (T) Rubber, Fuel Resistant, Non-Chromated Corrosion Inhibiting for Intermittent Use to 360 °F (182 °C)
SAE AMS3269	-	Sealing Compound, Polysulfide (T) Synthetic Rubber for Integral Fuel Tank and Fuel Cell Cavities High Strength, for Intermittent Use to 360 °F (182 °C) (Stabilized Type)
SAE AMS3276	-	Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)
SAE AMS3277	-	Sealing Compound, Polythioether Rubber Fast Curing for Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)
SAE AMS3278	-	Sealing and Coating Compound, Polyurethane (PUR) Fuel Resistant High Tensile Strength/Elongation for Integral Fuel Tanks/Fuel Cavities/General Purpose
SAE AMS3279	-	Sealing Compound, Sprayable for Integral Fuel Tanks and Fuel Cell Cavities, For Intermittent Use to 350 °F (177 °C) (Stabilized Type)
SAE AMS3281	-	Sealing Compound, Polysulfide (T) Synthetic Rubber for Integral Fuel Tank and Fuel Cell Cavities Low Density for Intermittent Use to 360 °F (182 °C)
SAE AMS3284	-	Sealing Compound, Low Adhesion, for Removable Panels and Fuel Tank Inspection Plates
SAE AMS3374	-	Sealing Compound Aircraft Firewall, Silicone
SAE AMS-C-83231	-	Coatings, Polyurethane, Rain Erosion Resistant for Exterior Aircraft and Missile Plastic Parts
SAE AMS-S-8802	-	Sealing Compound, Fuel Resistant, Integral Fuel Tanks and Fuel Cell Cavities
SAE AMS-S-83318	-	Sealing Compound, Polysulfide Type, Low Temperature Curing, Quick Repair, Integral Fuel Tanks and Fuel Cell Cavities
SAE ARP1110	-	Minimizing Stress Corrosion Cracking in Heat Treatable Wrought Low Alloy and Martensitic Corrosion Resistant Steels
SAE ARP4118	-	Isolation and Corrosion Protection of Dissimilar Materials, Carbon Composite Structure Repair
SAE AS6870	-	Nondestructive Inspection Program Requirements for Aerospace Systems

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SAE AS12500	-	Corrosion Prevention and Deterioration Control in Electronic Components and Assemblies
SAE AS81550	-	Insulating Compound, Electrical, Embedding, Reversion Resistant Silicone

(Copies of these documents are available online at https://www.sae.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC CPC-1/ NACE-21412-2020	-	Corrosion Prevention and Control Planning

(Copies of this document are available online at https://www.ampp.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Additive manufacturing (AM) processes. AM processes are processes to join materials to make parts from three-dimensional model data, usually layer by layer, also known as three-dimensional printing. These processes include, but are not limited to, processes such as powder bed fusion, fused filament fabrication, and directed energy deposition used to produce components or portions of components. For the purposes of this standard, processes for repair or geometry restoration such as cold spray processes, multi-pass robotic welding, etc. are included due to similarity of qualification approach. AM does not include coating processes such as those in accordance with MIL-DTL-5002.

3.2 Aerospace weapons systems. Applicable systems include fixed wing aircraft, unmanned aerial vehicles, space vehicles, rotorcraft, missile systems, air delivered guided munitions, armament systems and all related ground support equipment.

3.3 Aircraft structure. The structure of an aircraft includes the fuselage, wing, empennage, landing gear, rotorcraft rotor and drive systems, propellers, control systems and surfaces, airframe-engine interface components (including engine mounts, nacelles, air induction components), weapon mounts, structural operating mechanisms, components that perform a structural function, and other components as described in the contract specification.

3.4 Cognizant Engineering Authority (CEA). Cognizant Engineering Authority is an entity from the procuring activity that is responsible for the design of the weapons system and reviews and approves any deviations from the technical requirements contained within this standard.

3.5 Corrosion. The deterioration of a material or its properties due to a reaction of the material with its environment. Corrosion includes degradation of non-metallic materials. Some non-traditional examples include rotting of wood, degradation of concrete (carbonation, alkali-

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silica reaction phenomena), and degradation of composite materials due to reaction with the environment.

3.6 Corrosion prevention and control (CPC). The rigorous application of engineering, design and analysis, risk management, quality assurance, nondestructive inspection, manufacturing, operations, and support technologies to:

a. prevent the start of corrosion
b. minimize functional impairment from corrosion, and
c. define processes for the managing, tracking, and repair of corrosion problems.

3.7 Corrosion prevention and control plan (CPCP). This plan is approved by the CEA and describes the specific corrosion prevention and control measures to be implemented for the purpose of minimizing corrosion throughout the acquisition and sustainment lifecycle, as well as a plan for establishing a CPCP timeline (see Appendix A) and a Corrosion Prevention Team (see Appendix B).

3.8 Corrosion prevention and control planning. Consists of planning and establishing a management structure for CPC and the technical considerations and requirements in order to implement an effective CPC regime throughout the life cycle of a program. The key elements of CPC planning can be found in the Association for Materials Protection and Performance's (AMPP) Corrosion Prevention and Control Planning Standard (SSPC CPC-1/NACE-21412-2020) and in the Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment dated August 2022. A CPCP formally documents the CPC planning and execution, and is updated, refined, and matured as the program proceeds through the life cycle phases.

3.9 Corrosion prevention team (CPT). A team that consists of Government and contractor personnel within each aerospace weapons system management function with the purpose of focusing on and emphasizing CPC planning and execution throughout the life of the program. Examples include a Corrosion Prevention Advisory Board (CPAB), Corrosion Action Teams (CAT), etc.

3.10 Critical. For the purpose of this standard, the term "critical" is used to define a safety-of-flight (or fracture critical/failure critical) part, surface, or application on an aerospace weapons system that has been identified through analysis, test, or service history, the failure of which could have a catastrophic effect upon the aerospace weapons systems, and for which critical characteristics have been identified that are controlled to ensure the required level of integrity.

3.11 Damage. The presence of any crack, flaw, corrosion, disbond, delamination, or other feature that degrades, or has the potential to degrade, the performance of the affected component.

3.12 Durability. The ability of the structure or parts to resist cracking, corrosion, thermal degradation, delamination, wear, and the effects of foreign object damage for a prescribed period of time.

3.13 Erosion. The progressive loss of material from a surface due to impinging fluid or solid particles.

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3.14 Exterior surfaces. All surfaces normally exposed to an external environment are considered exterior surfaces. All interior surfaces that may also become wetted with water or corrosive fluid are considered exterior surfaces. These surfaces include, but are not limited to, wheels and landing gear, wheel wells and their fairings, control surfaces, wing-fold areas, battery compartments, and bilge areas on aircraft with latrines.

3.15 Materials compatibility. The resistance of a material to corrode, degrade (embrittlement, softening, etc.) or stain when it comes in contact with other chemicals or materials.

3.16 Microbially influenced corrosion (MIC). Corrosion that is caused or promoted by microorganisms such as bacteria or fungus. It can apply to both metals and non-metals.

3.17 Nondestructive inspection (NDI). The inspection of a structure or component in any manner that will not impair its future usefulness. The purpose of NDI may be to detect discontinuities at or beneath the external surface of a part, measure geometric characteristics, determine material structure or composition, or it may characterize physical, electrical, or thermal properties without causing changes in the part.

3.18 Stainless steels. Stainless steels are iron based alloys with a minimum of 10.5 percent chromium and are sometimes referred to as corrosion resistant steels (CRES). Stainless steels are generally classified into 5 groups: austenitic, martensitic, precipitation hardening, ferritic and duplex. Ferritic and duplex stainless steels are not generally used in aerospace weapons systems and therefore are not included in this standard.

3.19 Stress corrosion cracking (SCC). Cracking induced from the combined influences of sustained tensile stress and a corrosive environment on a susceptible material. SCC cracks grow at unpredictable rates and pose a flight safety risk.

3.20 Structural integrity. The condition that exists when a structure is sound and unimpaired in providing the desired level of structural safety, performance, durability, and supportability.

3.21 Supportability. Acceptable quality and cost-effective preventive methods, or in-service repair methods, are either available or can be developed in a timely manner to mitigate the deterioration of materials and structures fabricated using the selected manufacturing processes and joining methods.

3.22 Sustainment. Sustainment involves the supportability of fielded systems and their subsequent life cycle product support from initial procurement, to supply management (including maintenance), to reutilization and disposal. It can include functions such as initial provisioning, cataloging, inventory management and warehousing, and depot and field level maintenance.

3.23 Wear. Wear is the damaging, gradual removal or deforemation of a material at solid surfaces. Causes of wear can be mechanical (e.g., erosion) or chemical (e.g., corrosion).

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4. GENERAL REQUIREMENTS

4.1 Corrosion prevention and control plan (CPCP). The CPCP shall address the design, materials, and processes intended to be used on the specific system being procured including system interfaces, installation of government furnished equipment, and commercial items procured off-the-shelf. The key elements of CPC planning can be found in the Association for Materials Protection and Performance's (AMPP) Corrosion Prevention and Control Planning Standard (SSPC CPC-1/NACE-21412-2020) and in the Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment. The CPCP shall contain the following:

a. A combination of system design, engineering and manufacturing requirements, operations, logistics, and sustainment phases that mitigate corrosion throughout system life.

b. An integrated management structure that ensures ongoing, effective CPC communication and coordination among team members by defining roles and responsibilities for quality assurance, process control, materials/process engineering, manufacturing, low observables, technical writing, and Environment, Safety, and Occupational Health (ESOH) compliance.

c. Materials, processing methods, manufacturing techniques, and fabricating processes as well as the protective treatments identified in the finish specification.

d. Methods for CPC sustainability, logistics support, maintenance planning (such as methods/equipment for corrosion monitoring and non-destructive inspection), ground support and test equipment, handling, transportation, and storage of parts or assemblies, personnel training and the generation/validation/verification of technical information.

e. Performance requirements and verification methods for full stack-ups of the overall protective schemes, including low observable materials.

f. Methods to pursue compromises between conflicting requirements of corrosion mitigation, low observables, electrical grounding, environmental compliance, safety and occupational health. Life cycle costs shall be a factor in achieving compromises.

g. A requirements flow down process for suppliers, subcontractors, and sub-systems manufacturers and the procurement of commercial-off-the-shelf items that validates, verifies, and ensures their compliance with the CPC plan and finish specification.

h. Include an assessment of the planned basing and operational environments in which the system is intended to be used.

i. References to applicable specifications and standards.

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4.2 Finish specification. The finish specification shall be in accordance with MIL-STD-7179 using surface treatments defined in MIL-DTL-5002, unless otherwise approved by the CEA. These documents identify the specific materials, process methods, manufacturing techniques, protective pretreatments and treatments, organic and inorganic coatings, seals and sealants to be used for protection against corrosion of the assemblies, sub-systems and components selected for the specific system being procured. All finishing materials and processes shall conform to applicable federal and state regulations. Armament protection requirements shall be in accordance with MIL-STD-186.

4.3 Sustainment. Since corrosion prevention, control, and mitigation contribute significantly to the total cost of aerospace weapons systems ownership, the requirements of this standard shall be met during upgrades, procurement of spare parts, major repairs, modifications and similar support phases of existing systems. Department of Defense Directive (DoDD) 5000.01 recognizes that corrosion prevention and control is an issue over the life of an aerospace weapons system. DoDD 5000.01 emphasizes (1) the best time to off-set the effects of corrosion is early in system development, and (2) the requirements for the best design and selection of materials, processes and finishing systems needs to prevail during the operational and support phases of fielded aerospace weapons systems. Any procurement of spare parts, upgrades, repairs and other sustainment initiatives exceeding five million dollars shall require a CPCP and an accompanying process/finish specification, according to the Defense Federal Acquisition Regulation Supplement (DFARS) Part 207.105. The intent is to ensure coordination and implementation of the original CPCP and process/finish specification, that were approved during acquisition, be continued throughout system life. Elements of a sustainment corrosion program can be found in the “Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment.”

4.4 Material and process considerations in design.

4.4.1 Selection considerations. The primary consideration in the design and construction of aerospace weapons systems is the ability of the design to comply with structural and operational requirements. In addition, the aerospace weapons systems are expected to perform reliably and require minimum maintenance over a specified lifetime, which includes minimizing the rate of deterioration. Therefore, in the selection of materials and processing methods to satisfy system requirements, consideration shall also be given to those materials, processing methods and protective treatments that reduce failures due to deterioration. Corrosion-prone locations shall be identified by either validation testing as specified in MIL-STD-810 Method 509.8, Procedures II and III or by CEA approved modeling processes. Validation testing shall include selected materials, assembly techniques, and corrosion protection schemes in relevant environments and in-service loadings.

4.4.2 Guided missiles and rocket systems. Since guided missiles, rockets, armament, and their components have relatively low maintenance and high reliability performance requirements, MIL-STD-186, in conjunction with MIL-STD-1568, shall be used to establish materials and process requirements to protect these weapons systems from deterioration ensuring they meet their intended useful life.

4.4.3 Additive manufacturing processes. There are a multitude of AM processes for metallic

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applications, each with unique attributes. MIL-STD-1587 provides detailed requirements for additive manufacturing processes. The details of the AM process and feedstock used influence the microstructure including grain size, morphology and crystallographic texture, the size, distribution and morphology of porosity and voids, the surface topography, residual strain, and chemical in-homogeneities including the type and extent of phases present. Furthermore, post-build heat treatment operations can be employed to modify all of these attributes, and AM produced material may respond differently compared to classically produced materials under the same post-processing regimes. These attributes and resulting performance can vary between locations within the same part due to local processing differences. General statements about the corrosion performance of AM produced material are difficult to make and shall be considered on a case-by-case basis at this time.

4.5 General design requirements for corrosion prevention.

4.5.1 Exclusion of rain and airborne spray. The design of the system shall prevent water intrusion into, or being driven into, any part of the system interior or sealed components. All windows, doors, panels, canopies, etc. shall be provided with sealing arrangements that prevent entry of water when these items are correctly closed. Wetting of equipment, thermal insulation, and sound proofing materials shall be prevented. Recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors shall be provided for galleys, toilets, and crew compartments.

4.5.2 Ventilation. Ventilation to minimize moisture retention and buildup shall be included in the system design considerations and validated during system design testing.

4.5.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. Unless otherwise approved by the CEA, the minimum diameter for all drains shall be 0.375 inch (9.525 mm). All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to permit maximum drainage of accumulated fluids. Actual system configuration and attitude, in flight and on the ground, shall be considered in addition to component design. Drainage shall be provided for crew compartments, avionics bays and boxes, and engine bays/nacelles in such a manner that nearby structure shall not be compromised. Drain paths that are designed and incorporated into existing structure shall be appropriately noted and consideration given to ensuring paths are not obstructed due to manufacturing methods and materials (i.e. residual materials like sealants or corrosion preventive compounds). Additional action shall be focused on sustainment inspections to ensure all drain paths remain unobstructed and functioning as-designed. Areas that may be impractical to include drain paths (for example, hoist points) shall be noted and submitted to the CEA for approval. All drain hole locations and drain paths shall be documented and incorporated into the weapon system CPCP.

4.5.4 Galvanically dissimilar materials. Use of dissimilar material couples, as specified in MIL-STD-889, shall be avoided. When dissimilar material couples cannot be avoided, the dissimilar material couple shall be protected against galvanic corrosion by the specific protection schemes in accordance with MIL-STD-889 or by other methods authorized by the CEA.

NOTE: Galvanically dissimilar materials and the applicable corrosion mitigation

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strategy shall be identified, assessed, and evaluated during system design and throughout the system life-cycle. Testing of surrogate or actual systems that represent the final assembly may be required to quantify the corrosion risk and validate the mitigation strategy.

4.5.5 Insulation and acoustic blankets. When thermal, acoustical or other special purpose insulation blankets are used, they shall have a permanently bonded water repellent binder system applied or protected with sealant to prevent moisture from being absorbed into the blanket. Thermal blankets, for solid rocket motor stages and the post-boost propulsion system, are exempt from this requirement. Blankets shall be attached to the structure by a means that leaves sufficient space between the blanket and the underlying structure that avoids creating a crevice and permits drainage of moisture. The attachment design shall permit the blankets to be easily removed and re-attached to facilitate maintenance and inspection. Specific requirements shall be provided by the CEA.

4.5.6 Nondestructive inspection (NDI) for corrosion. For corrosion prone areas requiring inspection, access for inspection throughout the system lifecycle shall be evaluated during system design in order to identify inspectable and non-inspectable areas. Areas deemed non-inspectable may require additional CPC and prevention measures in order to mitigate the risk of not being able to inspect periodically. Coating selection and corrosion removal processes require consideration of their effects on NDI methods. SAE AS6870 shall be used to determine the key elements of NDI. When industry standard NDI methods cannot be utilized, alternate NDI methods may be utilized with the permission of the CEA.

4.6 Metallic materials. Restricted and prohibited metallic material, heat treatments and product forms are specified in MIL-STD-1587. The restricted or prohibited metallic materials specified in MIL-STD-1587 are based on low SCC resistance, low corrosion resistance, low fracture toughness, or other properties where there are new and improved metallic materials that offer better performance, improved flight safety and reduced maintenance requirements. The SCC resistance of weldments varies with the alloy and procedure used and shall be assessed in the procedure qualification record. Aerospace weapons systems parts that are currently in-service (i.e. legacy aircraft using traditional metals and alloys, heat treatments and product forms) are not restricted or prohibited for procurement of spares or replacement parts where there are no new drawing changes or modifications and where in-service data shows acceptable performance and low risk for continued use.

Many coatings, platings and other surface treatments used for corrosion protection of metals and alloys have the potential to reduce fatigue properties (for example, hard anodize coatings on aluminum alloys, engineering hard chrome plate on high strength steels, etc.). The design and selection of alloys and protective coatings shall take into account these potential reductions in fatigue properties. Shot peening or other residual compression stress inducing treatments shall be utilized to counter act these fatigue debits unless the fatigue debit is accounted for in the design. Compressive residual stress shall be applied via shot peening in accordance with MIL-STD-1587, or other methods approved by the CEA. Inclusion of the fatigue benefit of shot peening or other residual compression stress inducing treatments in design shall be approved by the CEA. Critical surfaces of all structural forgings, machined plate and extrusions, where accessible after final machining and heat treatment, shall be shot

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peened in accordance with SAE AMS2430 or SAE AMS2432, or other methods as approved by the CEA. Material removal limitations and maximum temperature limits following peening shall be in accordance with SAE AMS2430 and SAE AMS2432.

In addition to galvanic corrosion, there are other chemical reactions between metals and alloys that may cause chemical and metallurgical reactions that result in deterioration, amalgamation, embrittlement and premature failure. These reactions may occur at sub-ambient, ambient or elevated temperatures and do not necessarily require an electrolyte (for example, amalgamation between the metals). There are combinations of metals and alloys in aerospace where contact between the two can result in liquid metal embrittlement (LME) or solid metal embrittlement (SME). These issues are addressed in detail in accordance with MIL-STD-1587.

4.6.1 Aluminum.

4.6.1.1 Alloy selection. The selection of aluminum alloys for structural applications shall have a minimum SCC threshold of 25 ksi as tested in accordance with ASTM G44 and ASTM G47. All SCC testing shall be performed with test specimens in the same final heat treat condition of the representative part or assembly. If no post-production aging treatment is applied, a sufficient amount of time (not less than 2 weeks) shall be allowed for the material to stabilize. MIL-STD-1587 shall be used for restrictions on specific aluminum alloys and temper conditions due to their susceptibility to corrosion.

Relative SCC ratings for high-strength aluminum alloy products as specified in ASTM G47 and ASTM G64 are provided in table I. Although the ratings are based primarily on the results of standard corrosion tests, such as ASTM G64, an experience factor may be substituted for those materials that have established service records. To the extent possible based on product form, the ratings are given for tests conducted in the short transverse (ST) grain direction as this is the most critical SCC condition.

4.6.1.2 Aluminum alloy selection limitations. Cladding shall be used on 2000 and 7000 series aluminum alloys for external and interior aircraft surfaces in corrosive environments unless otherwise authorized by the CEA. These alloys shall be clad on both sides except where the design requires surface metal removal by machining or chemical milling. Cladding is not required under adhesively bonded surfaces. Clad aluminum alloy sheet shall be used in the following specific applications unless otherwise authorized by the CEA:

a. exterior skin that is 0.125 inch (3.175 mm) or less in thickness
b. parts that form a leading-edge
c. exhaust trail area of any source or wheel well area
d. parts that are resistance welded (spot or seam)
e. face sheets in bonded sandwich constructions

Non-clad materials may be used for the aileron skins, the flap shroud skins and the flap shroud closure pockets.

4.6.1.3 Compressive residual surface stress for corrosion resistance and fatigue life improvement. Surfaces of all structural forgings, machined plate and extrusions, where accessible

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after final machining and heat treatment, shall be shot peened to impart compressive residual stresses in accordance with MIL-STD-1587. Exceptions are components having web areas under 0.080 inch (2.032 mm) thick where no short transverse grain is exposed by machining. Those areas of forgings requiring lapped, honed, or polished surface finishes for functional engineering requirements shall be shot peened prior to performing these surface finishing operations.

4.6.1.4 Stress corrosion factor. Stress relief treatment and optimum grain flow orientation shall be used to minimize the hazard of SCC.

4.6.1.5 Aluminum lithium (Al-Li) alloys (special considerations). Initial Al-Li alloys had concerns for low thermal stability, high anisotropy of properties of semi-finished products (especially rolled ones), susceptibility to local deformation, high scattering of the values of fracture toughness, and deviation of the trajectory of fatigue cracks from the straightforward direction. While newer Al-Li alloys have addressed these concerns, Al-Li alloys shall only be used with permission from the CEA.

4.6.2 Steels.

4.6.2.1 High strength steels. All steel parts 180 ksi (1241 MPa) ultimate tensile strength (UTS) and above, including fasteners, require corrosion preventative treatments in accordance with MIL-DTL-5002 and MIL-STD-7179. Corrosion preventive treatments shall be proven to be non-embrittling in service environments in accordance with ASTM F519 testing using the 1A specimen, unless an alternate specimen is approved by the CEA. Elimination of hydrogen embrittlement (de-embrittlement) of high strength steels following coating deposition, if required, shall be accomplished by low temperature heating (baking) of the part or component in air for specific time periods in accordance with SAE AMS2759/9 or MIL-DTL-5002.

4.6.2.2 Stress corrosion factors. Press or shrink fits, taper pins, clevis joints in which tightening of the bolt imposes a bending load on the female lugs and straightening or assembly operations that result in significant sustained residual surface tensile stresses shall be avoided for alloy steel parts heat treated to 180 ksi (1241 MPa) UTS and above. In cases where such practices cannot be avoided, protective treatments, such as stress relief heat treatments, optimum grain-flow orientation, wet installed (with a protective material) inserts and pins and shot peening or similar surface working to minimize the hazard of SCC or hydrogen embrittlement damage, shall be applied. For steel weldments, the SCC resistance shall be prepared and tested in the final heat treatment condition in accordance with ASTM G58.

4.6.2.3 Stainless steels. When passivation is required by design, all stainless steels shall be passivated in accordance with SAE AMS2700, or equivalent, as approved by the CEA. While SAE AMS2700 permits the use of citric acid for passivation, use of this process shall be approved by the CEA. Table II provides relative corrosion resistance (general and SCC) for stainless steels in 3.5 percent NaCl exposure. Other stainless steel alloys shall have their SCC resistance determined by the agency authorized test methods and approved by the CEA. Stainless steels come in a wide variety of compositions, microstructures, and chemical and mechanical properties and find broad applications in aerospace weapons systems. The selection and use of stainless steels shall meet all design requirements (strength, weight, toughness, fatigue properties, corrosion resistance, etc.). Depending on specific design requirements, stainless steels shall be

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considered for applications in areas where severe corrosion conditions are expected, as an improvement over other alloys where corrosion performance has been poor and where use of stainless steel would preclude the need for corrosion protective surface finishes, coatings, or treatments that require the use of environmentally hazardous materials. Stainless steel alloys can cause or suffer from galvanic corrosion if coupled to dissimilar materials. MIL-STD-889 provides requirements and guidance on dissimilar materials and corrosion protection schemes to use to prevent galvanic corrosion. Additional restrictions and special precautions for the selection and use of stainless steels in aerospace weapons systems as specified in MIL-STD-1587.

4.6.2.3.1 Austenitic stainless steel. Austenitic stainless steels include the type 300 series alloys (301, 302, 304, 316, etc.). These alloys generally have very high corrosion resistance, especially in marine or seawater environments. If cold worked, the lower grade 300 series stainless steels, type 301 or 302 for example, can result in conversion of some of the austenite to ferrite and that can reduce corrosion resistance and cause the material to become slightly ferromagnetic. Sensitization of type 300 series is the precipitation of chromium carbide preferentially along the grain boundaries resulting in significant loss of corrosion resistance, especially SCC resistance. Sensitization of type 300 series stainless steels occurs by exposure to or slow cooling through the temperature range of 930 - 1562 °F (500 - 850 °C). Sensitization of stainless steels shall be avoided. Prevention of sensitization in austenitic stainless steels can be prevented by avoiding exposure to temperatures in the sensitizing range. Sensitization in austenitic stainless steels can be reversed by re-solution heat treating and rapid cooling. Use of stabilized austenitic stainless steel grades, where alloying elements, such as titanium (type 321) and niobium (type 347) that have a greater affinity for forming carbides than does chromium, are added to the chemical composition is another way to prevent loss of corrosion resistance. Reducing the carbon content to very low levels (grades designated with the letter “L”, such as 304L and 316L) is another way to prevent sensitization. Stabilized stainless steels are used when welding or brazing is required and re-solution heat treatment is not possible. Welding of un-stabilized type 300 series shall be followed by re-solution heat treatment. MIL-STD-1587 provides more detailed requirements and restrictions for welding and brazing of stainless steels. Free machining austenitic stainless steels, such as type 303 (sulfur addition) and type 303Se (selenium addition) are available where improved machinability is required. There is some drop off in corrosion resistance of these alloys compared to type 304 in severe sodium chloride environments, but their use is normally in small detailed machine elements within a larger system and not directly exposed to external environments.

4.6.2.3.2 Martensitic stainless steel. The type 400 martensitic stainless steels may be considered as high chromium counterparts to the quenched and tempered high strength steels. The most common 400 series stainless steels used in aerospace weapons systems are types 410 440A, 440B and 440C. Heat treatment (hardening) of these alloys involve a high temperature to form an austenitic structure (austenitizing) followed by quenching to form a hard and brittle martensite and then tempering to restore toughness and reduce hardness. When used in corrosive environments, type 400 series stainless steels shall have corrosion protective coatings and finishes applied. Tempering of 400 series martensitic stainless steels between 700 - 1100 °F (371 - 593 °C) shall be avoided because the corrosion and SCC resistance is reduced.

4.6.2.3.3 Precipitation hardening (PH) stainless steel. MIL-STD-1587 shall be used for

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specific restrictions and prohibitions on the use of PH stainless steels in specific heat treat conditions. PH stainless steels are classified by microstructure into three sub groups; martensitic PH stainless (e.g. 17-4 PH, 15-5PH) semi-austenitic PH stainless steels (e.g. PH 17-7, PH 15-7, PH, AM-355) and austenitic (A-286). The martensitic PH alloys (not to be confused with the martensitic stainless steels, Type 400 series) transform to a martensitic structure upon cooling from the solution heat treatment, however this martensite is very low in carbon content and is soft and ductile. Strength and hardness is increased by precipitation hardening (aging). The semi-austenitic PH stainless steels contain both a martensitic and austenitic microstructure when cooled from solution heat treatment as its chromium-nickel ratio prevents the formation of the fully austenitic phase. These alloys are then precipitation hardened to develop the final mechanical and physical properties. In general, the SCC resistance of PH stainless steels decreases as precipitation hardening temperatures decrease. These alloys are then precipitation hardened to develop the final mechanical and physical properties. MSFC-STD-3029 and SAE ARP1110 provide SCC resistance for PH stainless steels in various heat treated conditions (temper conditions) in sodium chloride environments broken down into high, moderate and low SCC resistance based on days to failure under sustained loading at 75 and 50 percent of yield strength using ASTM SCC test methods. MIL-STD-1587 provides specific restrictions and prohibitions on the use of PH stainless steels in specific heat treat conditions.

4.6.3 Titanium.

4.6.3.1 Corrosion and stress corrosion cracking resistance. The selection of titanium alloys for corrosive environments shall be based upon the corrosion resistance of the alloy and heat treatment condition in the specific environment, identification of total sustained stress (applied and residual), the correlation between laboratory test data and unique design requirements and in-service performance data. Titanium and its alloys generally provide excellent resistance to general corrosion, pitting corrosion and SCC in many applications including sodium chloride (marine) environments though there are exceptions for some alloys under some conditions. The tenacious nascent oxide film that forms instantly upon exposure to air is a significant factor in the overall excellent corrosion resistance. Commercially pure (CP) and many titanium alloys are virtually immune to SCC unless the titanium oxide is damaged or removed and the protective oxide cannot immediately re-form due to absence or exclusion of air (oxygen). SCC can occur in titanium alloys due to exposure to NaCl at both ambient and elevated temperatures (hot salt SCC) under some conditions. Alpha-Beta alloys, such as 6Al-4V, 3Al-2.5V, 5Al-2.5Sn, are rated as “high resistance” to 3.5 percent NaCl at room temperature under the testing conditions and rating criteria of MSFC-STD-3029. However, other titanium alloys such as 8Al-1Mo-1V and 6Al-2Sn-4Zr-2Mo have low resistance to NaCl exposure. Because of the significant effect of the protective titanium oxide film and whether it is present (or can reform) it can be difficult to compare laboratory SCC test results to in-service performance. Compressive surface stress processes (i.e. shot peening) shall be considered in certain cases where it is beneficial for the application. The appropriate standard test method for SCC resistance determination, as specified in ASTM G39, ASTM G44, ASTM G49, ASTM F945 shall be used and approved by the CEA when existing data is inadequate.

4.6.3.2 Surface contamination and cleaning. The surfaces of titanium products (sheet, plate, bar, forging, casting, metal additive, and extrusion) shall be 100 percent machined, chemically

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milled, or pickled to remove all contamination caused by oxidation and the formation of an alpha case formed while the material was at an elevated temperature in the presence of atmospheres containing oxygen. This includes contamination resulting from mill processing, heat treatment or elevated temperature forming operations. Removal of alpha case or other detrimental surface layers shall be confirmed by inspection techniques approved by the CEA. Welding of titanium alloys require special procedures to prevent oxygen absorption and embrittlement as specified in MIL-STD-1587. Cleaning of titanium parts for removal of organic and inorganic contaminates shall be performed in accordance with MIL-DTL-5002, or equivalent, as approved by the CEA. Care shall be taken to ensure that cleaning fluids and other chemicals are not used on titanium assemblies where entrapment can occur. Substances that are known to be contaminants and produce SCC include, but are not limited to, hydrochloric acid, trichloroethylene, trichloroethane, carbon tetrachloride, all chlorides, chlorinated cutting oil, halogenated hydrocarbons, fluorinated hydrocarbons, and methyl alcohol. All cleaning fluids and other chemicals as specified in MIL-DTL-5002, that are used during manufacturing and processing of titanium hardware for aircraft hardware shall be evaluated for potential corrosion or SCC susceptibility. Refer to relevant ASTM standards (i.e., ASTM G39, ASTM G44, ASTM G49, ASTM F945) for specimen preparation for SCC evaluation. ASTM F945 shall be used to test new or alternative cleaners for cleaning titanium alloy parts used at elevated temperatures (for example, compressor rotors, compressor blades, etc.).

4.6.3.3 Titanium wear. Titanium and its alloys exhibit very poor wear resistance and fretting can contribute to crack initiation, especially fatigue initiation, when the protective oxide layer is disrupted and corrosion occurs. If use of titanium in critical applications is required and fretting is likely, anodizing, in accordance with SAE AMS2488 or other wear-resistance materials, shall be considered. The effect of anti-galling, anti-fretting compounds on the corrosion performance of the joint shall be assessed by the CEA.

4.6.4 Other alloys.

4.6.4.1 Magnesium alloys. The selection and use of magnesium alloys for new designs for aerospace weapons systems shall be based only on the strength to weight ratio requirements utilizing the best corrosion resistant alloys and protective surface finishes and treatments as specified in MIL-DTL-5002 and MIL-STD-7179. Magnesium alloys are prohibited from use on Navy and Marine Corps aerospace weapons systems unless specifically authorized by the CEA due to historically significant corrosion problems with their use. Corrosion resistance shall be assessed using simulated corrosion testing and the similarity between the new design and successful prior applications (in-service experience). Due to having the lowest standard potential of all engineering metals, Magnesium and its alloys are highly susceptible to galvanic corrosion. MIL-STD-889 shall be used to determine galvanic compatibility for couples involving magnesium alloys and recommendations for the corrosion protection systems for galvanic couples involving magnesium alloys.

4.6.4.2 Copper Alloys used in structural applications. MIL-STD-889, MIL-DTL-5002 and MIL-STD-7179 shall be used for requirements and recommendations for corrosion protection of these materials. Bronzes (in both cast and wrought forms) such as aluminum bronze, nickel aluminum bronze, silicon bronze, manganese bronze, etc. and the copper beryllium alloys

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CDA 170, 172 and 173 have been extensively and successfully used for bushings and bearings in military landing gear shock struts cylinders, pistons, drag struts, actuators, etc. These copper alloys generally have acceptable SCC resistance in these applications where direct and severe exposure to NaCl environments occurs. Galvanic corrosion between these copper alloys and landing gear steels can be mitigated by application of surface treatments or electroplating to the faying surfaces of one or both components and use of sealants on the edges to exclude moisture (electrolyte).

4.6.4.3 Nickel-based alloys and cobalt-based alloys. MIL-STD-1587 shall be used for restrictions on these alloys. MIL-STD-889, MIL-DTL-5002 and MIL-STD-7179 shall be used for requirements and recommendations for corrosion protection of these materials. Nickel based alloys (e.g., Inconels, Incoloys, Hastelloys, etc.) and cobalt based alloys (e.g., Stellites, L-605, etc.) used in aerospace weapons systems are predominately for parts and components used in intermediate to high temperature service environments (engine components such as compressor, turbine and afterburner parts) where oxidation resistance, high temperature strength and creep resistance are required. For the most part, both of these alloy families have excellent corrosion and SCC resistance in NaCl (marine environments). At high temperatures, nickel based alloys are susceptible to sulfidation and sulfur embrittlement. Protection of these nickel based alloys to sulfidation frequently requires special (often proprietary) high temperature diffusion coatings.

4.7 Organic materials.

4.7.1 Organic material restrictions. The following restrictions shall apply to composites, elastomers, plastics, and other organic materials used in the design, fabrication and storage of aerospace structures and components:

a. Imide-based or graphite composites shall not be in contact with parts or materials that are susceptible to corrosion (aluminum, steel, tin) including, brackets, clips, gang channels, tubing, fasteners, etc. Protection requirements for composite and metal joints are specified in MIL-STD-7179.
b. All organic materials shall have resistance to degradation and aging (including resistance to hydrolysis, ozonolysis and other chemical processes exacerbated by atmospheric or operational exposure) consistent with performance requirements for the intended use.
c. Hazardous decomposition and other volatile, leachable out-gassing products that can cause corrosion when released by organic materials under operating conditions shall not come in contact with material components.
d. Cellular plastics, foams and wood shall not be used for skin stabilization in structural components, other than in non-metallic sandwich components. Due to its propensity to retain water, foam as sandwich core material shall not be used unless approved by the CEA.
e. Natural leather degrades quickly and shall not be used.
f. Elastomeric encapsulating compounds used shall conform to MIL-PRF-8516, MIL-PRF-23586, MIL-M-24041, MIL-A-46146, or SAE AS81550. Use of hydrolytically unstable encapsulation materials is prohibited.

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4.8 Other corrosion factors.

4.8.1 Corrosion prevention during manufacturing operations and storage. Precautions shall be taken during manufacturing operations and long-term storage of materials and commodities to prevent the introduction of corrosion or corrosive elements (see 4.8.5 and 4.8.6).

4.8.2 Cleaning. Cleaning of various types of metallic surfaces, prior to application of the surface treatments, inspections and coatings, shall be in accordance with MIL-DTL-5002 and TT-C-490, using materials and processes that have no damaging effect on the metal, including pitting, intergranular attack and significant etching. After cleaning, all parts shall be completely free of corrosion products, scale, paint, grease, oil, flux, and other foreign materials, including other metals, and shall be given the necessary treatment immediately after cleaning. Particular care shall be exercised in the handling of parts to ensure that foreign metals are not inadvertently transferred, which can occur when steel is allowed to come into contact with zinc surfaces and cadmium in contact with titanium. All closed compartments shall be cleaned prior to closeout assembly to remove debris such as metal chips, broken fasteners, dust, and other foreign material. Particular attention shall be given to ensure that drain holes are not blocked.

4.8.3 Surface damage. Damage to surface treatments or protective finishes shall be repaired prior to component assembly, including faying surfaces. Inorganic and organic coatings used for repair shall be the same as those on the undamaged areas or as approved by the CEA. Application of inorganic coatings shall be in accordance with MIL-DTL-5002. Application of organic coatings shall be in accordance with MIL-DTL-18264 (or MIL-DTL-53072 for Army Systems) using approved material specifications and qualified materials.

4.8.4 Marking pencils. Marking devices that contain graphite (for example, lead pencils) shall not be used to mark metal parts.

4.8.5 Protection of parts during storage and shipment. All parts and assemblies shall be given protection to prevent corrosion and physical damage during storage and shipment. Packaging practices shall conform to MIL-STD-2073-1.

4.8.6 Storage and shipping preservatives. MIL-PRF-46170 shall not be used as a rust inhibiting storage fluid. While MIL-PRF-46170 may prevent rust, it will deposit a barium containing compound onto metallic surfaces leading to potential component (e.g., valves) failures. Operational fluids, as specified in MIL-PRF-83282 and MIL-PRF-87257, shall be used to fill components instead of using MIL-PRF-46170. MIL-PRF-81322 shall not be used as a shipping preservative. MIL-PRF-32014 shall be used in place of MIL-PRF-81322 as a shipping preservative with few exceptions (see 4.10.3).

4.8.7 Inorganic finishes. Cleaning, surface treatments, and inorganic finishes for metallic surfaces of aerospace weapons systems shall be in accordance with MIL-DTL-5002. Any deviations from the detailed surface treatment and inorganic coating requirements as specified in MIL-DTL-5002, shall be approved by the CEA. Approval for new finishes shall be based upon detailed test plans that are formulated and executed to subsequently demonstrate equivalency (or better) to the inorganic finish being replaced.

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4.8.8 Organic finishes. All organic finishes and coatings shall be in accordance with MIL-STD-7179. Any deviations to MIL-STD-7179 materials and processes shall be approved by the CEA.

4.8.8.1 Organic finishes and systems (for Air Force systems). For Air Force weapons systems, all exterior coatings will use colors in accordance with SAE AMS-STD-595 and meet any operational requirements (i.e., thermal, low IR). For non-specialty coatings systems, a typical organic coating stack up for the inner mold line (IML) or the outer mold line (OML) shall be qualified conforming to MIL-PRF-23377 or MIL-PRF-85582 primer, any type or class as determined by the CEA, followed by MIL-PRF-85285 type IV topcoat. Additionally, an OML coating system qualified conforming to MIL-PRF-32239 may be used. Exterior surfaces subjected during service to high stress, structural flexing, or susceptible to extreme corrosive attack shall be sealed with MIL-PRF-81733, type III, or primed with TT-P-2760 followed by MIL-PRF-85285, type IV topcoat. In high temperature areas, the selected material shall be approved by the CEA. Those non-metallic components coated for erosion resistance with SAE AMS-C-83231, class A or B and which may be chemically stripped during sustainment shall have a wash primer base coat, applied prior to application of the class A or B polyurethane coats.

4.8.9 Environmental sealing. All permanent joints and seams located in exterior or internal corrosive environments, including those in landing gear wells, control surface vents, attachment wells and structure under fairings shall be faying surface sealed with sealants conforming to MIL-PRF-81733, SAE AMS3265, SAE AMS-S-8802, SAE AMS3281, SAE AMS3276, SAE AMS3277 or SAE AMS1389. Sealants used in integral fuel tanks shall conform to MIL-STD-7179 and include SAE AMS3278, SAE AMS3269, SAE AMS3279, SAE AMS-S-8802, SAE AMS3281, SAE AMS3276, SAE AMS3277, SAE-AMS-S-83318 or SAE AMS1389. When operational temperatures are anticipated to exceed 225 °F (107 °C) but not exceed 325 °F (163 °C) except in limited excursions to not more than 360 °F (182 °C), sealants conforming to SAE AMS3276, SAE AMS3277, SAE AMS3281 or SAE AMS1389 shall be used. For sealing areas that operate at 275 °F to 500 °F (135 °C to 260 °C), sealants conforming to SAE AMS3374 shall be used. Removable panels and access doors shall be sealed, either by rubber gaskets, molded seals, mechanical seals, or separable low-adhesion faying surface sealants conforming to SAE AMS3284. High adhesion sealants such as those conforming to SAE AMS-S-8802, SAE AMS3277, SAE AMS3281 or SAE AMS3276 may be used for access door sealing provided the proper parting agent is used on one surface. Alternative faying surface products, such as non-curing sealants and gel tapes and new or alternative technologies shall be approved by the CEA.

4.9 Fasteners.

4.9.1 Fastener installation. Permanent installations shall use MIL-PRF-23377 and removable installations shall use MIL-PRF-81733 or SAE AMS8802. For Naval Aviation Systems, both permanent and removable installations shall use MIL-PRF-81733 Grade A, unless an alternative is approved by the CEA. Waterborne primers, such as those in accordance with MIL-PRF-85582, shall not be used for permanent fastener wet installation. Sealant or corrosion inhibiting coatings other than those specified in MIL-STD-7179 shall not be used unless approval has been obtained from the CEA. Quick release fasteners and removable fasteners

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penetrating exterior surfaces shall be designed and installed so as to provide a seal to prevent moisture or fluids from entering. Holes for these fasteners shall be primed in accordance with MIL-STD-7179 or MIL-PRF-23377, type I, class C1 or C2 epoxy primer and allowed to completely cure prior to installing the fastener. These fasteners shall be installed with corrosion preventive compound. Maintenance instructions shall require periodic reapplication of this material in-service.

4.9.2 Fasteners in titanium. Titanium, Monel, and stainless steel fasteners installed in titanium structures may be installed dry, unless sealing is required for liquid tightness or pressurization. Cadmium plated interference fit fasteners shall not be used in contact with titanium.

4.9.3 Fasteners in graphite composites. Fastener materials for use in carbon fiber reinforced composite structures shall be titanium, A286, or Inconel 718. Cadmium plated fasteners and aluminum fasteners shall not be used.

4.9.4 Monel and stainless steel fasteners. Monel fasteners or stainless steel fasteners shall be coated with low hydrogen embrittlement (LHE) Zn-Ni, cadmium, or aluminum in accordance with MIL-DTL-5002 when used in contact with aluminum components. With prior CEA approval, uncoated Monel and stainless steel fasteners may be used if both wet installed with primer or sealant and encapsulated with sealant.

4.9.5 Titanium fasteners. Titanium fasteners in contact with aluminum shall be wet installed. For maximum corrosion protection, titanium fasteners should be coated with LHE Zn-Ni, class 3, type II, in accordance with MIL-DTL-32648 and wet installed, in accordance with MIL-STD-7179.

4.9.6 Interference fit fasteners. Fastener holes for interference fit fasteners shall be primed with MIL-PRF-23377, type I, class C1 or C2 or MIL-PRF-85582, type I, class C2 and shall be completely dry prior to assembly. Primer formulations containing water are prohibited for wet installation of fasteners. Wet installation may be waived when using pre-coated fasteners, if approved by the CEA.

4.10 Special considerations.

4.10.1 Signature control. Low observable materials, processes, primary and secondary structures used to achieve signature control shall be evaluated in accordance with SAE ARP4118 and MIL-STD-889 for corrosion resistance in full stack-up representative assemblies.

4.10.2 Fluids, lubricants and their containment. Only qualified military specification or government-adopted industry specification materials shall be used for hydraulic fluids, coolants, greases, gas turbine oils, and other functional fluids. Phosphate ester based hydraulic fluids (such as Skydrol®) are not active and shall not be used for new design unless approved by CEA, typically for commercially derivative aircraft.

4.10.3 Greases. Design shall ensure that incompatible greases are not mixed during assembly. MIL-PRF-32014, a corrosion-resistant lithium soap grease, shall be used in lieu of

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non-corrosion inhibited greases in applications where the operating temperature is less than 350 °F (176 °C), unless an alternative grease is approved by the CEA. If deemed necessary by the CEA, MIL-PRF-27617 shall be used in lieu of MIL-PRF-81322. For gaseous oxygen compatibility requirements or high temperature applications, perfluorinated greases in accordance with MIL-PRF-27617, shall be used. Greases in accordance with MIL-PRF-32014 can be used above 350 °F (176 °C), if the performance is evaluated with component level testing and approved by the CEA. For launch vehicle only, perfluoropolyether (PFPE) greases are permitted, however they shall not be used for actively moving hardware during flight. A change from polyalphaolephin (PAO) or mineral oil based grease to a perfluoro-polyalkylether (PFPAE) grease shall occur during component replacement at depot to ensure adequate cleaning and that incompatible greases are not mixed.

4.10.4 Coolant. For radar cooling systems, PAO based coolant in accordance with MIL-PRF-87252 or equivalent with CEA approval shall be used.

4.10.5 Electronic or avionics systems. All electronic assemblies, electrical wiring inter-connect systems (EWIS) and avionics shall be in accordance with SAE AS12500.

4.10.6 Firefighting agents. Fire extinguishing agents such as fire fighting foam (FFF), bromochloromethane, dibromochloromethane or dry powder agents such as potassium bicarbonate or sodium bicarbonate after exposure to heat are corrosive. Decontamination procedures require flushing with generous quantities of water in conjunction with washing and rinsing of all surfaces and components exposed to fire suppressant materials. To the extent possible, designs shall minimize areas of potential exposure to these materials and facilitate the flushing and cleaning of those areas exposed. If fire retardants may not be fully removed from equipment such as aircraft engines, compatibility of the retardants with various alloy families need to be evaluated to minimize potential damage of these components.

4.10.7 Wear and erosion. The design and manufacture of aircraft shall include practices to minimize damage by wear and erosion. Coatings for the prevention of wear and erosion shall be in accordance with MIL-STD-7179.

4.10.8 Lubrication. Provisions shall be made for lubrication of all parts subject to wear using lubricants from MIL-HDBK-275 as specified in MIL-HDBK-838. The fire resistant synthetic hydrocarbon hydraulic fluid in accordance with MIL-PRF-83282 shall be used as the aircraft hydraulic fluid in operating temperatures down to -40 °F (-40 °C). For operating temperatures down to -65 °F (-54 °C), MIL-PRF-87257 shall be used. Multipurpose lubricants, such as the wide temperature general purpose grease described in 4.10.3, shall be used whenever possible, without compromising performance and reliability. Components that are highly loaded, dynamic and in potentially corrosive environments (for example, landing gear, arresting gear) shall make maximum use of lubrication fittings, versus other forms of lubricants. Graphite containing lubricants shall not be used in contact with aluminum, ferrous, magnesium, zinc, or cadmium alloys or platings, since they may promote corrosion of these metals. Graphite containing lubricants may be safely used in contact with corrosion resistant stainless steels, titanium, nickel alloys, and cobalt alloys, and similar corrosion resistant metals and alloys.

4.10.9 Verification of corrosion design. Aerospace weapons systems are usually designed

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for specified lifetimes with maintenance according to defined maintenance concepts and plans. Specific corrosion testing shall be conducted on components and subsystems in accordance with MIL-STD-810 Method 509.8. Testing shall identify corrosion risks and mitigation to meet anticipated performance requirements as determined by the CEA. Additionally for naval systems, these assemblies shall be tested to 500 hours of ASTM G85.A4 and finish and corrosion protection verification criteria shall be in accordance with Appendix C3.5.

4.10.10 Engine corrosion testing. Engine materials and coatings shall be corrosion tested under simulated engine environmental conditions commensurate with their final usage during operation, handling, and storage of the engine. For this testing, a new or newly overhauled engine shall be used. Prior to starting the test, the engine shall be disassembled and an inspection shall be conducted to determine the condition of all parts exposed to atmospheric conditions. Detailed photographic coverage of these parts shall be used for comparison with post-test conditions. The engine shall then be reassembled, pretest performance shall be calibrated, and subject to 25 accelerated mission test (AMT) cycles (see JSSG-2007 for more information on AMT cycles) while being injected with a two percent of airflow weight spray solution, consisting of the following materials dissolved with distilled water to make one liter of salt spray solution:

Chemical designation	Quantity per liter of spray solution
NaC1 (c.p.)	23 grams
$Na_2SO_4 \cdot 10H_2O$	8 grams
Stock solution (See recipe below)	20 milliliters

The stock solution shall be composed of the following materials dissolved with distilled water to make one liter of stock solution:

Chemical designation	Quantity per liter of stock solution
KC1 (c.p.)	10 grams
KBr	45 grams
$MgC1_2 \cdot 6H$ O (c.p.)	550 grams
$CaCl_2 \cdot 6H$ O (c.p.)	110 grams

At specified intervals during the test, the engine shall be subjected to internal inspections to detect any evidence of corrosion or progression of corrosion on internal parts. Upon completion of the test, a performance check shall be conducted and the engine disassembled and inspected for evidence of corrosion. Detailed photographs shall be taken of all parts that show evidence of corrosion. The test results shall be considered satisfactory when the extent of corrosion is not of such a magnitude as to impair structural integrity or component operation, or be a cause of significantly reducing performance, engine durability, or parts.

5. DETAIL REQUIREMENTS (Not applicable.)

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to provide Department of Defense Program Offices with a document that provides timely and comprehensive consideration during systems design of corrosion prevention and control processes and of the lessons learned over the years from operational systems worldwide. System reliability and maintainability will be significantly improved by the use of this standard. It should be used in accordance with MIL-STD-1587 in selection of materials and processes that will meet the requirements of the systems being designed in accordance with MIL-STD-1530 and MIL-STD-3063.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Corrosion Prevention and Control Plan (CPCP) (see 4.1).

c. Finish specification (see 4.2).

d. If Nondestructive Inspection (NDI) is required (see 4.5.6).

e. Timelines and milestones for the CPCP and finish specification (see Appendix A).

f. The establishment of Corrosion Prevention Teams (CPTs) (see Appendix B).

NOTE: Corrosion Prevention Teams are referred to by many different names depending on the Service such as Corrosion Action Teams (CAT), Corrosion Prevention Action Teams (CPAT), Corrosion Working Groups (CWG), etc. For example, the Air Force refers to these teams as Corrosion Prevention Advisory Boards (CPABs).

6.3 Associated Data Item Descriptions (DIDs). This standard has been assigned an Acquisition Management Systems Control (AMSC) number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DIDs must be listed on the Contract Data Requirements List (DD Form 1423).

DID Number	DID Title
DI-MFFP-81402	Finish Specification
DI-MFFP-81403	Corrosion Prevention and Control Plan (CPCP)

The above DIDs were current as of the date of this standard. The ASSIST database should be researched at https://quicksearch.dla.mil to ensure that only current and approved DIDs are cited on the DD Form 1423.

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6.4 Subject term (key word) listing.
Metal finishes
Organic coatings
Sealants
Stress corrosion

6.5 Alternate materials and processes. Justification data, including both laboratory and service experience, will be required for CEA approval of any alternate materials and processes.

6.6 Document websites and historical information websites. Relevant documents and historical information can be found by accessing the links below:

https://www.dau.edu/tools/t/CPC-Guidebook

https://www.dscc.dla.mil/Programs/MilSpec/DocSearch.aspx

https://www.quicksearch.dla.mil

www.grantadesign.com (Cambridge Material Selector)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE I. Rating for resistance to SCC aluminum alloys in the short transverse (ST) grain direction(1).

Alloy and Temper	Rolled plate	Rod and bar	Extruded Shapes	Forgings	General Rating According to Al Standards (2)
2014-T3, T4, T451	-	-	-	-	Intermediate
2014-T6	Low	Low	Low	Low	
2024-T3, T4	Low	Low	Low	-	
2024-T6	-	High	-	Low	
2024-T8	High	Very High	High	Intermediate	
2025-T6	-	-	-	-	Intermediate
2099-T83	-	-	47.8 KSI (LT)	-	
2099-T86	52.2 KSI	-	48.5 KSI (LT)	-	
2117-T4	-	-	-	-	Very High
2124-T8	High	-	-	-	
2124-T851	High (RLTYS (3))	-	-	-	
2218-T61	-	-	-	-	Intermediate
2218-T72	-	-	-	-	Intermediate
2219-T3, T37	Low	-	Low	-	
2219-T6	Very High	Very High	Very High	Very High	
2219- T8, T87	Very High	Very High	Very High	Very High	
2219- T851	Very High (RLTYS (3))	-	-	-	
2219-T8510, T8511	-	-	30 KSI	-	
2618-T61	-	-	-	-	Intermediate
6061-T6	Very High	Very High	Very High	Very High	
6063- T6	-	-	-	-	Very High
7005-T53, T63	-	-	Low	Low	
7039-T64	Low	-	Low	-	
7049-T73	Very High	-	High	Very High (RLTYS	
7049-T7352	-	-	-	Very High (RLTYS	
7049-T74	Very High	-	High	High	
7049-T76	-	-	Intermediate	-	

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TABLE I. Rating for resistance to SCC aluminum alloys in the short transverse (ST) grain direction(1) – Continued.

Alloy and Temper	Rolled plate	Rod and bar	Extruded Shapes	Forgings	General Rating According to Al Standards (2)
7050-T73510, T73511	-	-	45 KSI	-	
7050-T736	High	-	High	High	
7050-T74	High	-	High	35 KSI	
7050-T7451	35 KSI	-	-	-	
7050-T74510, T74511	-	-	35 KSI	-	
7050-T7452	-	-	-	35 KSI	
7050-T76	Intermediate	High	Intermediate	-	
7050-T7651	25 KSI	-	-	-	
7050-T76510, T76511	-	-	17 KSI	-	
7075-T6	Low	Low	Low	Low	
7075-T73	Very High	Very High (RLYS (3))	Very High	42 KSI	
7075-T7351	Very High (RLTYS (3))	Very High (RLTYS (3))	Very High	-	
7075-T73510, T73511	-	-	Very High (RLYS (3))	-	
7075-T7352	-	-	-	Very High (RLTYS	
7075-T736	-	-	-	High	
7075-T74	-	-	-	35 KSI for thickness 3 in. and less. High RLYS (3) for thickness	
7075-T7452	-	-	-	35 KSI for thickness = or < 3 in. High RLYS (3) for thickness > 3 in.	
7075-T76	Intermediate	-	25 KSI	-	
7075-T7651	25 KSI	-	-	-	

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TABLE I. Rating for resistance to SCC aluminum alloys in the short transverse (ST) grain direction(1) – Continued.

Alloy and Temper	Rolled plate	Rod and bar	Extruded Shapes	Forgings	General Rating According to Al Standards (2)
7075-T76510, T76511	-	-	25 KSI	-	
7085-T7452	-	-	-	35 KSI	
7079-T6	Low	-	Low	Low	
7149-T73	-	-	High	Very High	
7149-T74	-	-	High	High	
7175-T736	-	-	-	High	
7175-T74	-	-	-	35 KSI	
7175-T7452	-	-	-	35 KSI	
7175-T7454	-	-	-	35 KSI	
7178-T6	Low	-	Low	-	
7178-T76	Intermediate	-	Intermediate	-	
7475-T6	Low	-	-	-	
7475-T73	Very High	-	-	-	
7475-T7351	40 KSI	-	-	-	
7475-T76	Intermediate	-	-	-	
7475-T7651	25 KSI	-	-	-	

(1) Aluminum stress-corrosion cracking ratings are sourced from ASTM G64, "Aluminum Standards and Data 2017", MIL-STD-1568D, and ALCOA Green Letters. Ratings are found in the short transverse grain direction (ST), unless specified otherwise (e.g. long transverse or LT).
   a. Per ASTM G44 and ASTM G47, alternate immersion stress corrosion tests in 3.5 percent sodium chloride were completed.
   b. Ratings based off ASTM G64,
      i. Very High—Equal to or greater than 75 percent of the specified minimum yield strength.
      ii. High—Equal to or greater than 50 percent of the specified minimum yield strength.
      iii. Intermediate—Equal to or greater than 25 percent of the specified minimum yield strength or 100 MPa (14.5 KSI), whichever is higher.
      iv. Low—Fails to meet the criterion for rating C.

(2) Per "Aluminum Standards and Data 2017",
   a. "These ratings are neither product specific nor test direction specific and therefore indicate only the general level of stress-corrosion cracking resistance. For more specific information on certain alloys, see ASTM G64."

(3) RLTYS—Registered long transverse yield strength. RLYS—Registered longitudinal yield strength. Use these values in note (1) when specified for the material.

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TABLE II. Corrosion characteristics of stainless steels.

Class	Alloy	General Corrosion Resistance	Stress corrosion Resistance
Austenitic	301	High	Very high
	302	High	Very high
	304	High	Very high
	310	High	Very high
	316	Very high	Very high
	321	High	Very high
	347	High	Very high
	21-6-9	Moderate	Susceptibility varies significantly with composition, heat treatment, and product form.
Martensitic	440C	Low to moderate - will develop superficial rust film with atmospheric exposure	Susceptibility varies significantly with composition, heat treatment and product form
	420		
	410		
	416		
Precipitation Hardening	13-8Mo	Moderate	Susceptibility varies significantly with composition, heat treatment, and product form.
	15-7Mo	Moderate	
	14-8Mo	Moderate	
	17-4PH	Moderate	
	15-5PH	Moderate	
	AM355	Moderate	
	AM350	Moderate	
	9Ni 4Co-0.20C	Moderate	Very high
	9Ni 4Co-0.30C	Moderate	Very high
	9Ni 4Co-0.45C	Moderate	Low
Other	A286	High	Very high

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TIMELINE AND MILESTONES FOR THE CORROSION PREVENTION AND CONTROL PLAN (CPCP) AND FINISH SPECIFICATION

A.1 SCOPE

A.1.1 Scope. This appendix details the timeline and milestones for the submission of a corrosion prevention and control plan (CPCP) and finish specification (see 6.2e). This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

A.1.2 CPCP submission timeline. For major capability acquisition programs under DoDI 5000.85, the initial draft of the corrosion prevention and control plan should be submitted to the CEA as part of the proposal package 60 days prior to the System Requirements Review (SRR), Milestone A. The submittal of the corrosion prevention and control plan and the finish specification should occur 60 days prior to System Design Review (SDR), Milestone B. Follow-on revision and submittal of these documents should be accomplished 60 days prior to Preliminary Design Review (PDR), Critical Design Review (CDR) and as required to properly record a change to materials and processes being selected for corrosion prevention and control. Through design studies, analysis of failure reports and on-site inspections, data should be collected and analyzed for revisions to these documents. For sustainment initiatives, the CPCP and the finish specification should be adjusted accordingly and submitted 45 days after the procurement vehicle is signed.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in the Appendix of this standard. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they should meet all specified requirements of documents cited in Appendix A of this standard, whether or not they are listed.

A.2.2 Other Government documents, drawings, and publications. Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE DIRECTIVE

DoDI 5000.85	Major Capability Acquisition

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

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CORROSION PREVENTION TEAMS (CPTs)

B.1 SCOPE

B.1.1 Scope. This appendix details the procedure for establishing a corrosion prevention team (CPT) (see 6.2f) for Major Capability Acquisitions. This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

B.2 APPLICABLE DOCUMENTS

B.2.1 General. The documents listed in this section are specified in the Appendix of this standard. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they should meet all specified requirements of documents cited in Appendix B of this standard, whether or not they are listed.

B.2.2 Other Government documents, drawings, and publications. Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE DIRECTIVE

DoDI 5000.85	Major Capability Acquisition

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

B.3 PROCEDURES

B.3.1 Corrosion Prevention Team (CPT). A Government-only CPT should be established and appropriately resourced for Major Capability Acquisition Programs under DoDI 5000.85 during the conceptual design phase of the program prior to Milestone A outlined in Appendix A. This government-only team, designated by the program manager, provides technical support during proposal evaluation. After contract, a contractor-only CPT (also referred to as the Contractor Corrosion Team [CCT]) should be established and appropriately resourced to ensure that CPC requirements are being implemented during all phases of the aerospace weapons systems/equipment lifecycle.

B.3.2 Government and contractor membership. Each team should consist of, as a minimum, a Team Lead, and members from Program/Project Management, Systems Engineering, Life Cycle Logistics, Test and Evaluation, Facilities, Materials and Processes, Design, Quality Assurance, Manufacturing, Contracts, etc. Membership suggestions are in the "Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment." Each CPT is unique and is tailored several times during the life cycle of the system to satisfy the changing requirements of the program.

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B.4 DUTIES

B.4.1 Duties of the Corrosion Prevention Team (CPT). The team should:

- Define operating procedures, and prescribe appropriate specifications and standards to be used in the system/equipment.

- Outline or document the Request for Proposal (RFP) and contract requirements necessary to execute the CPCP.

- Address system/equipment definition, design, engineering development, production, and sustainment phases, ensuring they are consistent with the design life and affordability of the system/equipment.

- Develop options for finish and coating systems to be used on the procured system/equipment.

- Review all design considerations, material selections, costs, and documentation, including Statements of Work (SOWs) that may affect CPC throughout the life of the system and equipment.

- Advise the PM on corrosion-related issues, assess the adequacy of the corrosion maintenance plan, documentation and guidance as they are developed, and elevate any shortcomings or unresolved issues.

B.4.2 Duties of the Contractor Corrosion Team (CCT). The CCT should:

- Ensure that CPC requirements are being implemented during all phases of the aerospace weapons system being procured.

- Ensure appropriate documents outlined under the CPCP are prepared and submitted in accordance with the required schedule.

- Support any necessary design reviews, clarification, and resolution of any differences in technical position along with supporting final approval of the documentation on a timely basis.

The chairperson's duties include:

a. Serving on the CPT as required.
b. Establishing periodic meetings as required to resolve problems as they occur. Other meetings should be convened, as required, or when a critical or major problem arises that requires action by the team.
c. Signing off (or designee) on all production drawings after review of materials selection, treatments and finishes.
d. Maintaining a continuing record of all action items and their resolutions.
e. Establishing the principal tasks to be accomplished to implement CPC procedures in the contractor and subcontractor manufacturing facilities.

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B.5 Implementation of a Joint CPT. When required by the service program, the government program manager should establish and appropriately resource a CPT at the start of the Engineering Manufacturing Development Stage (Milestone B as specified in Appendix A). This joint Government and contractor team, co-chaired by both a Government and prime contractor representative, supports the program manager by addressing corrosion related issues, such as, design considerations, selection of materials and processes, program management, documentation, supportability, reliability, maintainability, sustainability and structural integrity. The CPT should meet on a regular basis in accordance with the established team charter throughout the life of the aerospace weapons system to ensure program continuity and should also be utilized during sustainment initiatives.

B.5.1 CPT membership. CPT membership should include authoritative representatives from appropriate Government and contractor organizations necessary to ensure that proper materials, processes, and treatments are selected and subsequently properly applied, and maintained from the initial phase through the life cycle of the system. Suggestions on membership are in the "Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment."

B.5.2 CPT charter. A charter should be established at the start of Milestone B after the CPT is organized. A charter consists of an introduction, purpose, membership (Government and contractor), responsibilities (Government and contractor), procedures and an appendix addressing the membership and contact information.

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MISCELLANEOUS FACTORS AFFECTING CORROSION

C.1 SCOPE

C.1.1 Scope. This appendix details miscellaneous factors that affect corrosion. This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

C.2 APPLICABLE DOCUMENTS

C.2.1 General. The documents listed in this section are specified in the Appendix of this standard. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they should meet all specified requirements of documents cited in Appendix C of this standard, whether or not they are listed.

C.2.2 Other Government documents, drawings, and publications. Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Engineering Circular EC-434-000-003	Material Selection Restrictions List for Navy and Marine Corps Aviation Weapons Systems
Engineering Circular EC-434-000-004	Risk Mitigation for Non-chromate Coating Systems
Engineering Circular EC-434-000-005	Corrosion Prevention and Control for Navy and Marine Corps Aviation Systems

(Copies of these documents are available from the following address: Commander, NAVAIRSYSCOMHQ, Air Vehicle Engineering Department, Materials Division, Building 2187 Suite 3340, 48110 Shaw Rd., NAS Patuxent River, MD 20670-1906)

DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT (DFARS)

Subpart 223	Environment, Energy And Water Efficiency, Renewable Energy Technologies, Occupational Safety, And Drug-Free Workplace
Subpart 252	Solicitation Provisions and Contract Clauses

(Copies of these documents are available at https://www.dcaa.mil/dfars.html.)

C.2.3 Non-Government publications. The following documents form a part of this

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document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D2247	-	Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D2803	-	Standard Guide for Testing Filiform Corrosion Resistance of Organic Coatings on Metal

(Copies of these documents are available at https://www.astm.org.)

C.3 MISCELLANEOUS INFORMATION

C.3.1 Microbially influenced corrosion (MIC). MIC can occur in a variety of locations such as fuel systems (i.e., tanks, pipelines) or areas that experience high humidity, standing water, entrapment of water, or pooled or leaked operational fluids. MIC may be indicated by the presence of "slime," sludge or fungus. Water or fuel that has a foul or acidic (e.g., vinegar) odor or a cloudy, flocculent appearance can also indicate the presence of microbes. MIC should be investigated in corrosion associated with water or fuel with these characteristics. If MIC is suspected, it should be reported to the fuel quality control agency for each branch of service (e.g., Air Force Petroleum Agency). Microbiologists should be consulted to help identify if MIC is present and what options can be used to mitigate the MIC.

C.3.2 Material restrictions for Navy and Marine Corps Aircraft. Engineering circular EC-434-000-003 provides information on the hazards or restrictions for the use of materials and processes, not only for environmental, safety, and health concerns, but also for minimizing adverse effects on systems due to poor design.

C.3.3 Hexavalent chromium regulations. Hexavalent chromium (Cr^{6+}) is a chemical that has been used for decades in numerous DoD weapons systems and platforms due to its corrosion protection properties. However, Cr^{6+} is a known carcinogen. Defense Federal Acquisition Regulation Supplement (DFARS) Subparts 223 and 252 mandate programs to minimize the use of materials containing Cr^{6+} in items acquired by DoD. It codifies a DoD policy for addressing the serious human health and environmental risks related to the use of Cr^{6+}. The rule prohibits the delivery of items containing more than 0.1 percent by weight Cr^{6+} in any homogeneous material under DoD contracts unless there is no acceptable alternative to the use of Cr^{6+}. The mandatory prohibition does not apply to legacy weapons systems (developed before 2012), unless they are undergoing a major modification. Even for new and modified systems, it is not intended to preclude the use of Cr^{6+} for critical defense applications for which there are no alternatives that can meet performance requirements. If the program manager determines that Cr^{6+} is necessary for the weapons system until viable alternatives are proven, the program may seek an Authorization Request for the contract. The authorization request is developed by the program's hazardous materials control and management expert (SME) in coordination with the

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corrosion subject matter expert (SME). The authorization should be approved by the Senior Approving Official in conjunction with the Military Department's Corrosion Control and Prevention Executive. Further guidance for Naval Programs is available in Appendix 1 of Engineering Circular EC-434-000-005 and Engineering Circular EC-434-000-004.

C.3.4 Biobased material selection. On 12 September 2022, the President signed an Executive Order 14081 on Advancing Biotechnology and Biomanufacturing for a Sustainable, Safe, and Secure American Bioeconomy (Ref 1). This order requires all procuring agencies to establish a biobased procurement program, the intent being to increase the usage of biobased materials by 2025. Additional corrosion testing may be necessary, depending on the product and application, to ensure materials that appear to be chemically similar to legacy materials match or exceed the legacy materials' corrosion prevention properties. Biobased alternatives to materials listed in MIL-STD-1568 include but are not limited to lubricants, coolants, coatings, elastomers, composites, and other polymers. Department of Defense Manufacturing Technology Program (Ref 2) manufacturing innovation institutes such as BioMADE (Ref 3) and their government program management team can assist with the development of biobased product procurement plans as well as assessing these materials for the intended applications.

Ref 1) https://www.federalregister.gov/documents/2022/09/15/2022-20167/advancing-biotechnology-and-biomanufacturing-innovation-for-a-sustainable-safe-and-secure-american
Ref 2) https://www.dodmantech.mil
Ref 3) https://www.biomade.org

C 3.5 Naval Systems design verification criteria. The follow design verification criteria are applicable for Naval Aerospace weapons systems:

Test	Criteria
ASTM D2247 30 day humidity test	No blistering, softening, loss of adhesion or other film defect
ASTM B117 2000 hours salt spray test with scribed panels	No blistering, lifting of coating nor substrate corrosion.
ASTM G85.A4 500 hours SO_2 salt spray test with scribed panels (Navy only)	No pitting greater than 1 millimeter in depth.
ASTM D2803 1000 hours filiform corrosion test with scribed panels.	No filiform corrosion extending beyond ¼ inch from the scribe.

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 20

Preparing activity:
Navy - AS
(Project MFFP-2023-002)

Review activities:
Army - AV
Navy - CH, EC, MC, OS, SA, SH, YD
Air Force - 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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